Unit 1: Ratios and Proportions

Topic 1: Equivalent Ratios — **Foundation Level 3** 🌶

Name: ____________________

Date: ____________________

Q1. If the ratio of x to y is 2:3, which pair (x, y) fits?

(A) (4, 9)
(B) (10, 15)
(C) (6, 8)
(D) (3, 2)

Q2. Complete the ratio table: 1:4, 2:8, 3:12, ?:20.

(A) 4
(B) 5
(C) 6
(D) 7

Q3. A recipe calls for 2 cups of sugar for every 5 cups of flour. How many cups of flour are needed for 10 cups of sugar?

(A) 20
(B) 25
(C) 15
(D) 50

Q4. Simplify the ratio 45:60 to its simplest form.

(A) 9:12
(B) 3:4
(C) 5:6
(D) 15:20

Q5. On a coordinate plane, if you plot the ratio 1:3, which point is equivalent?

(A) (2, 5)
(B) (3, 9)
(C) (4, 10)
(D) (5, 12)

Q6. Ratio of blue marbles to red is 7:2. If there are 14 red marbles, how many are blue?

(A) 49
(B) 28
(C) 21
(D) 35

Q7. A car travels 120 miles in 2 hours. What is the simplified ratio of miles to hours?

(A) 60:1
(B) 120:2
(C) 30:1
(D) 1:60

Q8. [**FRQ**] Create a ratio table for 3:5. Then, list the pairs as (x, y) coordinates for the first 3 rows.

Q9. [**FRQ**] If for every 5 votes Candidate A received, Candidate C received 3 votes, how many votes did Candidate C get if Candidate A got 50? Show your work.

Q10. [**FRQ**] Simplify 36:48 and 15:20. Are these ratios equivalent? Explain why or why not.

Chef's Tips

- **Plotting:** In a ratio $x : y$, x is the horizontal move and y is the vertical move on a graph.
- **Scale Factor:** To find the next row in a table, multiply the original ratio by the same whole number.
- **Total Quantities:** Sometimes you need to add the ratio parts (like $2 + 3 = 5$) to find a "total" relationship.